A Technical and Economic Assessment of Onboard Hydrogen Storage Systems for Light-Duty Vehicles

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Introduction and Objective

Our objective is to inform hydrogen storage R&D by answering:

- **What are the technical and economic barriers in H_2 storage preventing the commercialization of light-duty Fuel Cell Electric Vehicles?**
- **Which H_2 technology or technologies will be the most viable in the nearby and far futures?**
- **What is the recommended government R&D funding for advancing hydrogen storage systems?**

Identification of Metrics

- DOE highlights that "substantial improvements must be made in the weight, volume, and cost" (DOE, p. 9)
- Gravimetric Capacity = $\frac{\text{mass of usable } H_2}{\text{storage system's mass}}$
- Volumetric Capacity = $\frac{\text{mass of usable } H_2}{\text{storagesystem's volume}}$
- Cost = $\frac{\text{system's total production cost}}{\text{lower heating value of usable } H_2}$

Literature Review

- Sources selected based on author's credentials, publication date (since 2012), and relevance to evaluating metrics for H_2 storage systems
- 25 primary, secondary, and government sources

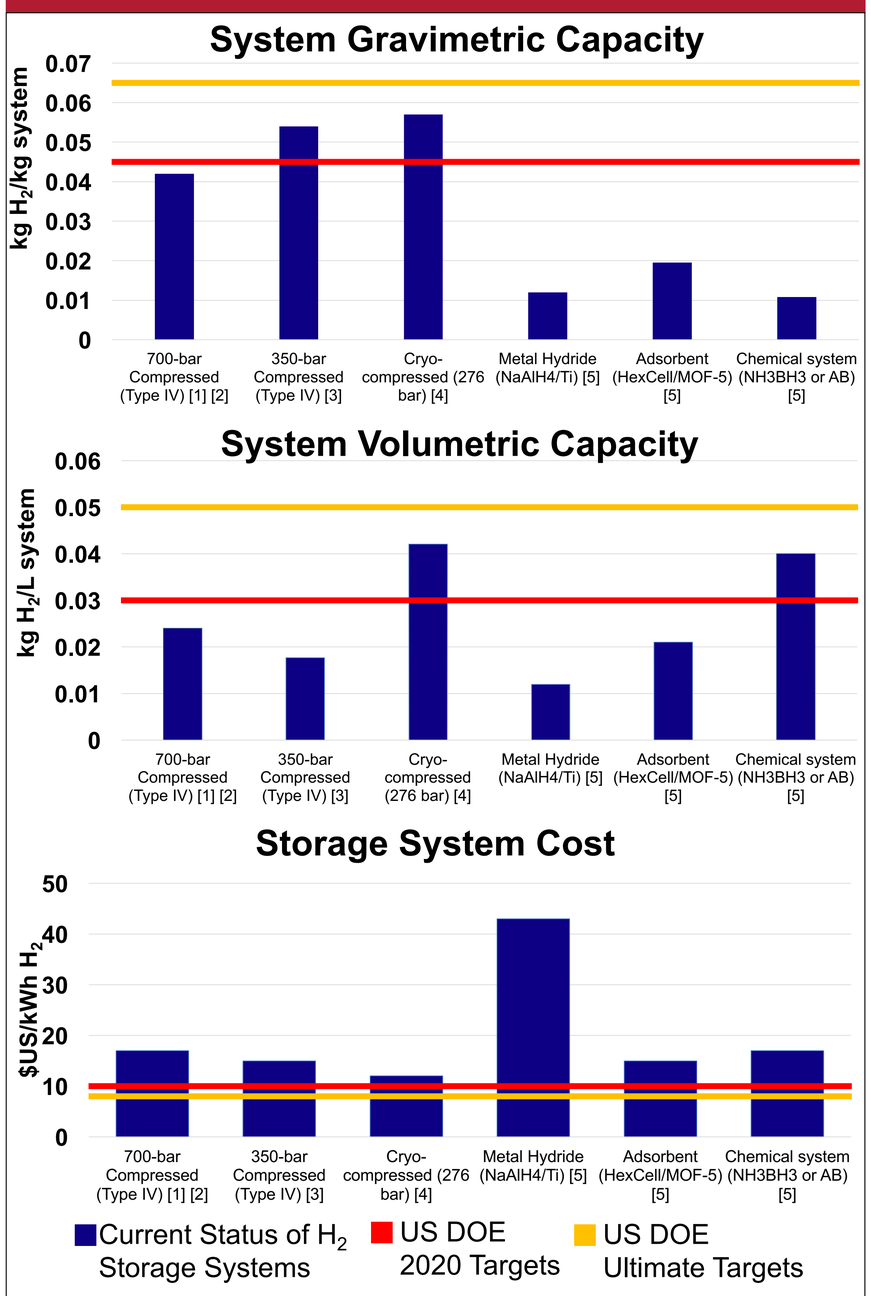
Expert Elicitation

- Experts selected based on education, work experience, patents, and publications
- 30 interviews (1-2 hrs) across academia, government, and industries
- Elicited experts' upper bound, lower bound, and best guess

Assumptions

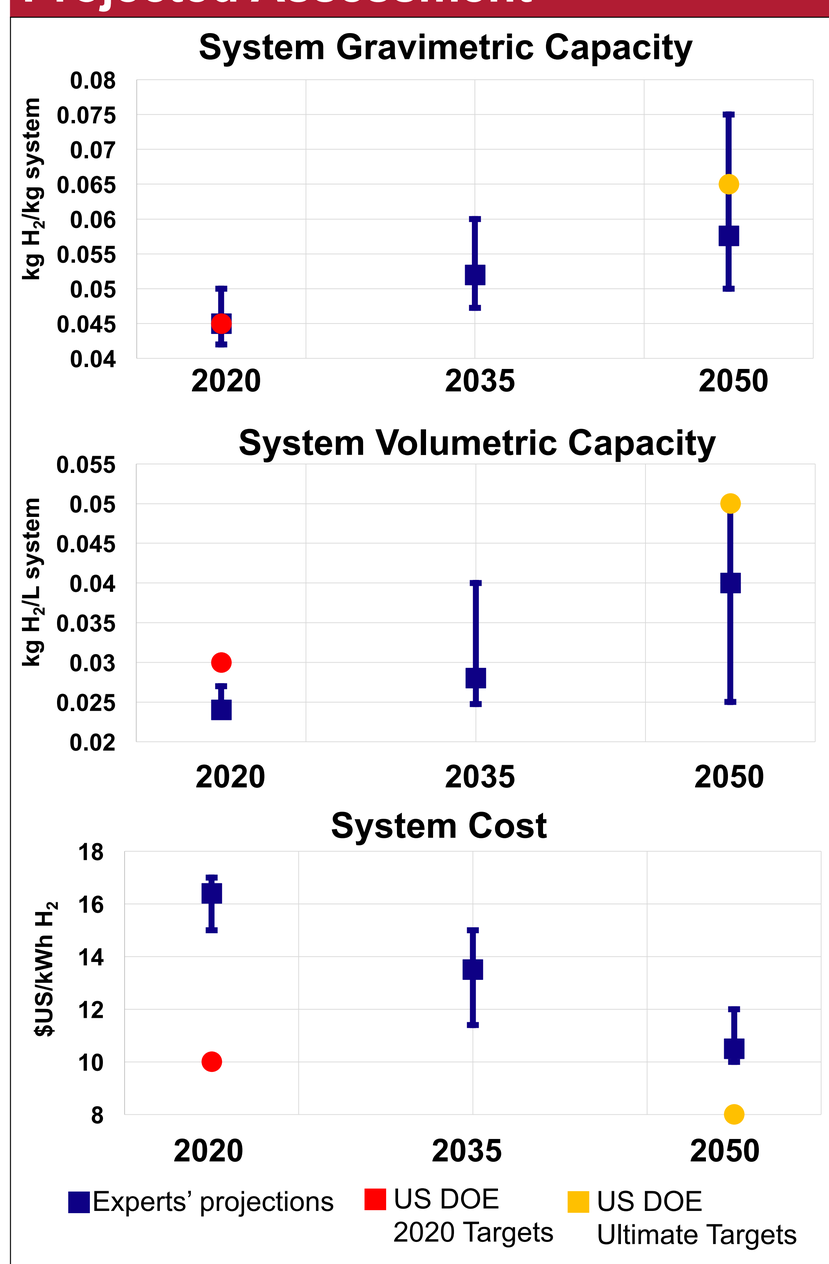
- Stores 5.6 kg of usable H_2 [2] [8]
- Endures 1,500 cycles, 5,000 hrs of operation, or 150,000 miles [7]
- 500,000 $\frac{\text{units}}{\text{year}}$ total production volume
- Cost excluded any sales markup applied by the final system assembler for profit, overhead, and other business expenses; cost could include markups applied by lower-tier suppliers

Current Assessment



- Cyro-compressed is the most competent system to meet the nearby targets for gravimetric capacity, volumetric capacity, and cost
- Compressed H_2 storage in carbon fiber wound Type III or Type IV pressure vessels cannot meet .04 kg H_2/L system at 350-700 bar [4]
- No metal hydride has had a capacity of at least 10 – 11% wt hydrogen with an enthalpy 25 – 27 kJ/mole H_2 to be technically viable [6]
- In simulations of 137,000 hypothetical MOF structures using grand canonical Monte Carlos, 30,000 MOFs structures are able to theoretically exceed 2025 DOE targets [7]

Projected Assessment



Experts' projections for H_2 storage technologies

Hydrogen Storage Systems	Year		
	2020	2035	2050
Compressed gas	22	14	9
Cold-Compressed	0	1	0
Cryo-Compressed	1	4	3
Metal Hydride	0	0	0
Adsorbent	0	4	4
Chemical	0	1	3
No technology specified	8	9	13

- Compressed gas is most technically and economically viable in 2020
- Adsorbent or cyro-compressed would be most viable in 2035 and 2050
- Experts recommend $19.5 million total funding ($10 million in material development, $5 million to advanced tanks, $2.5 million to testing and analysis, and $2 million to energy)

Summary

- Cyro-compressed is the most competent to meet nearby targets
- Compressed gas is the most viable in 2020 while material-based technologies could be more viable in the future
- Experts recommend $19.5 million in funding (DOE: $15.6 million)

Acknowledgement and References

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